

Experimental Components for PDF Processing System

This document outlines the various components where we can experiment with different models and approaches.

Component Experimentation Matrix

Phase	Component	Current Choice	Alternative Options	Key Considerations
Preparation	PDF Processing	Marker	PyMuPDF, PDFMiner, pdf2image + Tesseract	Text/image extraction quality, structure, speed
Preparation	Image Captioning	Florence-2	BLIP-2, GPT-4V, LLaVA, CogVLM	Caption accuracy, technical content, speed, resources
Preparation	Text Embedding	CLIP	Sentence Transformers, BERT variants, OpenAI embeddings	Embedding quality, semantic search, speed, cost
Preparation	Vector Database	ChromaDB	Pinecone, Weaviate, Qdrant, Milvus	Query performance, scalability, metadata, deployment
Preparation	Chunking Strategy	Section-based	Fixed-size, sentence-based, paragraph-based, ML-based	Context, search relevance, overhead, structure

Phase	Component	Current Choice	Alternative Options	Key Considerations
Preparation	Image Processing	Direct storage	Image compression, thumbnail, feature extraction	Storage, retrieval speed, quality, resource usage
Querying/Answering	Query Embedding	CLIP	OpenAI embeddings, Sentence Transformers, custom models	Query understanding, cross-modal, speed, cost
Querying/Answering	Retrieval	Vector similarity	Hybrid (BM25+vector), multi-stage, context-aware	Retrieval accuracy, speed, context, diversity
Querying/Answering	Reranking	None	Cross-encoder, custom rerankers, rule-based	Result quality, latency, cost
Querying/Answering	Answer Generation	LLM (e.g. GPT-4)	Llama, Mistral, Claude, custom fine-tuned models	Answer quality, context window, cost, latency

Notes

1. Each component can be evaluated based on:

- Performance metrics
- Resource requirements
- Integration complexity
- Maintenance overhead

2. Consider running A/B tests for:

- Different model combinations
- Various chunking strategies
- Alternative storage solutions
- Retrieval strategies

- Reranking approaches
- Answer generation models

3. Key metrics to track:

- Processing time
- Memory usage
- Storage requirements
- Search accuracy
- Caption quality
- Query response time
- Retrieval relevance
- End-to-end latency
- Answer quality

4. RAG-specific considerations:

- Context window management
- Query understanding accuracy
- Cross-modal retrieval effectiveness
- Result diversity and relevance
- System latency and throughput